# PRODUCT REQUIREMENTS DOCUMENT
# Industrial Predictive Maintenance & Safety Monitoring Node

### Smart Machine Health Monitor using ESP32 | Version 6.0 (True Final Freeze)

| | |
|---|---|
| **Course** | Logic Design & Microprocessors |
| **Platform** | ESP32 (PlatformIO preferred) + Antigravity Free |
| **Version** | 6.0 - Final Freeze |
| **Date** | 05 November 2026 |
| **Budget** | <= Rs. 2000 extra (target Rs. 720-1235 with contingency) |
| **Resources** | ESP32 + Arduino Kit + 3-4 Laptops |

**Purpose:** True final freeze. FR10 is present in the FR table, Appendix diagrams are complete with exit transitions, all typos are corrected, and minor implementation notes are included. Ready for implementation and academic submission.

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## 1. Overview & Problem Statement

Industrial machines fail due to bearing wear, overheating, overload or imbalance. This project implements a low-cost ESP32 Machine Health Monitoring Node that reads 3-4 sensors, applies filtered threshold logic via a finite-state machine with hysteresis and a safety-aware SILENCED flag, and provides local + multi-laptop web visualisation. Development uses Google Antigravity Free.

## 2. Assumptions & Constraints

- **Logic-level safety (critical):** DHT22 and SW-420 are powered at 3.3 V only. Digital outputs and pull-ups connect only to 3.3 V. ESP32 GPIOs are not 5 V tolerant.
- **ACS712 supply:** ACS712 is a 5 V device. Power it at 5 V. Its analog output (centred near 2.5 V at zero current, up to ~4.5 V at full scale) must pass through the 10kΩ + 15kΩ divider before reaching GPIO 34. Only the digital sensors and their pull-ups are at 3.3 V.
- **ACS712 preferred variant:** 5 A (sensitivity 185 mV/A). If 20 A is used, sensitivity = 100 mV/A; update scale factor in code.
- **DHT22 preferred:** requirement >= 2 s between reads; recommended implementation interval >= 2.5 s. DHT11 needs +1 s after power-up before the first valid read; discard the first reading after boot. If DHT11 is used, accuracy is lower (+/- 2 C) and range is 0-50 C; therefore cap the critical temperature threshold at <= 50 C.
- **OLED:** Monochrome 0.96-inch SSD1306. State shown by text/icon/inverted blocks. Colour only on web dashboard and status LED(s).
- **Status indication:** RGB LED or three discrete LEDs (green/yellow/red) preferred. If only a single LED is available, use the exact blink-rate patterns defined in FR5. LED pattern is driven by state only; SILENCED affects the buzzer only.
- **Vibration sensor:** Digital (SW-420/HW-608). Pot adjusted so idle = LOW and gentle taps produce clean pulses.
- **Optional 4th sensor (DS18B20):** Display/log-only unless a threshold is explicitly configured. Humidity is display/log-only.
- **Wi-Fi:** SoftAP in a typical classroom. SoftAP password is shown on OLED at boot and on a printed card.
- **Power source:** 5 V USB power bank (>= 1 A). No battery management in this version.
- **ADC1:** Used for the current sensor because Wi-Fi activity can interfere with ADC2.
- **Brownout detector:** Should be left enabled (default on most ESP32 Arduino/PlatformIO configs).
- **Thresholds:** All thresholds and confirmation times are named constants and will be calibrated on the actual hardware.

## 3. Goals & Objectives

- Correct sensor timing: DHT22 >= 2 s (recommended >= 2.5 s, discard first post-boot reading); other sensors 1-2 Hz via a simple scheduler.
- FSM with hysteresis, concrete vibration condition, and SILENCED escalation that clears SILENCED on WARNING->CRITICAL.
- Local monochrome OLED state indication + defined LED patterns + buzzer + debounced acknowledge.
- SoftAP (WPA2) web dashboard (SSE preferred) for 3-4 laptops.
- Non-blocking design + task watchdog for 45-min demo robustness.
- Demonstrate ADC (divider, attenuation, calibration formula), I2C, safe 3.3 V GPIO use, timers, sequential logic.


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- Majority of code/dashboard generated and refined with Antigravity Free (PlatformIO preferred).
- Extra hardware cost target Rs. 720-1235 (hard limit Rs. 2000).

## 4. Scope

### In Scope

- ESP32 node with Vibration, Temp+Humidity, Current (+ optional 4th display/log sensor).
- FSM with hysteresis + SILENCED flag (cleared on escalation).
- Monochrome OLED, LED behaviour, buzzer, debounced acknowledge.
- SoftAP + SSE/polling dashboard for 3-4 clients.
- Serial CSV logging; optional LittleFS with size limit.
- Watchdog, non-blocking loop, invalid-sensor detection (FR10), full test plan, this PRD.

### Out of Scope

- Paid cloud services, on-device ML, multi-node mesh, custom PCB, battery management.


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## 5. Functional Requirements (FR1-FR10)

| ID | Requirement | Priority |
|---|---|---|
| FR1 | Sample Vibration (digital, debounced) and Current (ACS712 after divider) at 1-2 Hz. Sample DHT22 at interval >= 2 s (recommended implementation >= 2.5 s, discard first reading after boot). Optional 4th sensor at 1-2 Hz. Implement a simple sensor scheduler. | Must |
| FR2 | Moving-average / exponential filter on analog channels. Software debounce 50-100 ms on vibration and on acknowledge button (GPIO 27). | Must |
| FR3 | FSM: NORMAL / WARNING / CRITICAL with hysteresis. SILENCED flag (buzzer off). On escalation WARNING > CRITICAL, clear SILENCED and re-enable buzzer (new acknowledge required). SILENCED also clears on return to NORMAL. LED pattern follows state only; SILENCED affects buzzer only. | Must |
| FR4 | 0.96-inch monochrome OLED shows live values + state text/icon (or inverted block). Colour is used only on the web dashboard and status LED(s). | Must |
| FR5 | LED behaviour: Prefer RGB or 3 discrete LEDs (G/Y/R). If only a single LED is available use exact patterns: NORMAL = off or 1 short blink every 2 s; WARNING = 2 blinks per second; CRITICAL = 5 blinks per second or solid ON. | Must |
| FR6 | SoftAP with WPA2 non-default password (shown on OLED at boot). Dashboard via SSE (preferred) or 1 s AJAX polling. Support 3-4 concurrent laptop clients. | Must |
| FR7 | Serial CSV: timestamp_ms, vibration, temp_C, humidity_pct, current_A, state, silenced. Optional LittleFS append with max-size or simple rotation. Default is Serial-only for demo. | Must |
| FR8 | Non-blocking main loop (millis() or FreeRTOS tasks). No long delay() calls that starve network or display. | Must |
| FR9 | Enable ESP32 task watchdog (or software watchdog) so a firmware hang recovers without manual reset during the 45-min demo. | Must |
| FR10 | Detect invalid/disconnected sensor readings (NaN, out-of-range, open-circuit) and indicate the error on OLED and dashboard. Remaining sensors continue to operate. | Must |

## 6. Non-Functional Requirements

- **Performance:** Analog/digital sensors 1-2 Hz; DHT22 >= 2 s (impl. >= 2.5 s). Dashboard latency < 1.5 s. State transition within configured confirmation window + 1 s.
- **Reliability:** 45 min continuous operation; watchdog recovers from hangs; non-blocking design; invalid-sensor detection (FR10).
- **Usability:** Dashboard on Chrome/Edge/Firefox. SSID/password on OLED + printed card.
- **Security:** WPA2 SoftAP, non-default password. No open network.
- **Cost:** Target Rs. 720-1235 (subtotal + 20-30% contingency). Hard limit Rs. 2000.
- **Power:** Peak ~300 mA; 5 V / 1 A power bank is sufficient. ACS712 quiescent current is approximately 10 mA at 5 V.
- **Safety:** Digital sensors and pull-ups at 3.3 V only. ACS712 at 5 V with divider.


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## 7. Hardware Specification & Exact BOM

### Already Available

ESP32 DevKit, breadboard, jumper wires, basic LEDs, push buttons, USB cable / power bank.

### Bill of Materials (extra components)

| Item | Qty | Approx. Rs. | Notes |
|---|---|---|---|
| Vibration SW-420 / HW-608 | 1 | 50-80 | Power at 3.3 V only |
| DHT22 (or DHT11) | 1 | 100-150 | Vcc = 3.3 V; 10 kOhm pull-up to 3.3 V |
| ACS712 5A (preferred) | 1 | 100-150 | Vcc = 5 V; 185 mV/A; output via divider |
| 0.96-inch OLED SSD1306 I2C | 1 | 180-250 | Monochrome |
| Active buzzer | 1 | 20-30 | HIGH = on |
| RGB LED or 3x LEDs (G/Y/R) | 1 set | 15-40 | Preferred; else single LED + blink patterns |
| 10 kOhm resistor | 2 | 2-5 | DHT22 pull-up + divider top |
| 15 kOhm resistor | 1 | 2-5 | Divider bottom (~3.0 V at 5 V in) |
| Optional DS18B20 | 0-1 | 80-120 | Needs 4.7 kOhm pull-up to 3.3 V |
| 4.7 kOhm resistor (opt.) | 0-1 | 2-5 | DS18B20 1-Wire pull-up |
| Protoboard + wires | 1 | 50-80 | Neater assembly |
| **Subtotal** | | **Rs. 600-950** | |
| **+ 20-30% Contingency** | | **Rs. 720-1235** | Still under Rs. 2000 |

**Voltage Divider (ACS712 OUT => ESP32 ADC):** 10 kOhm (from ACS712 OUT to ADC pin) + 15 kOhm (ADC pin to GND). Ratio = 15 / (10 + 15) = 0.6. A 5.0 V signal at the divider input appears as ~3.0 V at the ADC pin, safely inside the ADC attenuation 11 dB range (approximately 0-3.1 V). Use ADC1 (GPIO 34).


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## 8. Pin Mapping & Conflict Check

| Function | GPIO | Type | Notes / Conflict Check |
|---|---|---|---|
| DHT22 Data | 4 | IO | Vcc & pull-up = 3.3 V. Safe. |
| Vibration Digital | 25 | IO | Vcc = 3.3 V. Safe. Avoids strapping pins. |
| ACS712 (after divider) | 34 | Input-only | ADC1 (Wi-Fi safe). No internal pull-up. External divider only. |
| OLED SDA | 21 | IO | I2C. Safe. |
| OLED SCL | 22 | IO | I2C. Safe. |
| Buzzer | 13 | IO | Active buzzer (HIGH = on). Safe. |
| Acknowledge Button | 27 | IO | Internal pull-up, active LOW, 50-100 ms debounce. Safe. |
| Status LED / RGB | 26, 32, 33 | IO | Three safe outputs for R/G/B or single LED on GPIO 26. No strapping pins. |
| Optional DS18B20 | 14 | IO | 1-Wire + 4.7 kOhm pull-up to 3.3 V. Safe. Not shared with LEDs. |
| Avoid completely | 0, 2, 12, 15 | - | Strapping pins. Do not use for sensors, buttons or LEDs. |

No pin is assigned to more than one function. GPIO 12 and GPIO 15 are never used. GPIO 27 is exclusively the acknowledge button. Status LEDs use 26/32/33 only.

## 9. Software Architecture & Antigravity Workflow

**Environment:** PlatformIO inside Antigravity Free (preferred) or Arduino IDE + ESP32 core. Agents generate sensor skeletons, FSM, web UI and docs. Workflow: prompt => review/integrate => flash/test => refine prompt with observed errors => repeat.

**Modules:** Sensor Manager (scheduler, filter, debounce, ACS712 offset/scale, discard first DHT22 read, invalid detection FR10) | FSM (hysteresis + SILENCED, clear on escalation) | Display (OLED text/icon) | Network (SoftAP WPA2 + SSE) | Logger (Serial CSV + optional LittleFS size-capped) | Main (non-blocking, watchdog FR9).

## 10. Logic Design - Finite State Machine

States: NORMAL, WARNING, CRITICAL. Separate boolean SILENCED. Acknowledge sets SILENCED (buzzer off, visual state retained). Escalation rule: On transition WARNING -> CRITICAL, clear SILENCED and re-enable the buzzer. A new acknowledge is required if the operator wants silence again. SILENCED also clears automatically on return to NORMAL. *LED note: LED pattern is driven by the current state only; SILENCED affects the buzzer only.*


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### Concrete Conditions & Hysteresis (calibrate on hardware)

| Channel | Warning Entry | Critical Entry | Exit (hysteresis) |
|---|---|---|---|
| Vibration (debounced digital) | Digital HIGH continuously for > 2 s OR > 3 pulses in any 2 s window | Condition persists or intensifies for additional 2 s | LOW / quiet for 5 s |
| Temperature (DHT22) | > 45 C confirmed for 3 s | > 55 C confirmed for 2 s (if DHT11: <= 50 C) | < 42 C for 5 s |
| Current (ACS712 5A, 185 mV/A) | > 1.5 A confirmed for 2 s | > 2.5 A confirmed for 2 s | < 1.2 A for 5 s |
| Humidity / optional 4th | Display & log only (not in default alarm logic) | - | - |

All confirmation windows and thresholds are named constants. Success criteria use "within configured confirmation window + 1 s".

## 11. Data Logging, Dashboard & Security

**CSV columns:** timestamp_ms, vibration, temperature_C, humidity_pct, current_A, state, silenced
**Example:** `123456,1,28.4,55.1,0.85,WARNING,0`

**LittleFS (optional):** append-only with max file size (e.g. 64-128 kB) or simple rotation. Default = Serial only.

**Dashboard:** SSE preferred. SoftAP WPA2 non-default password shown on OLED at boot. SoftAP typical limit 4-8 clients; if unstable keep only Laptop 2 as primary dashboard.

## 12. Power Budget

| Component | Typical | Notes |
|---|---|---|
| ESP32 Wi-Fi active | 160-240 mA | Peak TX |
| OLED | 15-25 mA | - |
| Sensors combined | 5-15 mA | Digital at 3.3 V; ACS712 quiescent ~10 mA at 5 V |
| Buzzer + LED active | 25-40 mA | Only when alarming |
| **Peak estimate** | **~300 mA** | 5 V / 1 A bank sufficient |

## 13. Risk Register

| Risk | Impact | Mitigation |
|---|---|---|
| 5 V logic into ESP32 GPIO | High | Power DHT22 & SW-420 at 3.3 V only; pull-ups to 3.3 V. |
| ACS712 powered at 3.3 V by mistake | High | Explicit note: ACS712 Vcc = 5 V; output through divider only. |
| Strapping pin LOW at boot | High | Only safe GPIOs used (see pin table). GPIO 12/15 never used. |


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| Risk | Impact | Mitigation |
|---|---|---|
| ACS712 over-voltage at ADC | High | 10k+15k divider keeps ~3.0 V; 11 dB atten; verify with 5.0 V source. |
| False vibration triggers | Med | Debounce + 2 s / multi-pulse confirmation. |
| SoftAP client limit / interference | Med | Keep Laptop 2 primary; fallback OLED+Serial. |
| Hang during long demo | Med | Task watchdog (FR9) + non-blocking design. |
| DHT22 too-fast reads / disconnect | Med | Scheduler >= 2.5 s recommended; FR10 invalid/NaN detection. |


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## 14. Calibration Procedure & ADC Formula

### ADC / Current conversion (implementation note)

1. Read raw ADC (12-bit, attenuation 11 dB). Convert to voltage at the ADC pin.
2. Divider input voltage = ADC_pin_voltage * (10 + 15) / 15 = ADC_pin_voltage * 25/15.
3. Current (A) = (divider_input_voltage - zero_offset_voltage) / sensitivity
   where sensitivity = 0.185 V/A (5 A module) or 0.100 V/A (20 A module).
4. **Zero offset location:** Measure with no load current. Convert the ADC-pin reading back to divider-input voltage (step 2) and store that value as zero_offset_voltage (expected near 2.5 V). Alternatively store the ADC-pin zero and apply the same conversion consistently in the current formula. Do not mix the two domains.

- **Divider verification:** Apply a known 5.0 V source at the divider input (bench supply, not from the ACS712). Confirm ADC pin reads ~3.0 V and stays below 3.1 V.
- **ACS712 zero:** With no load, record average divider-input voltage; store as zero_offset_voltage.
- **DHT22:** Wait ~1 s after power-up, discard the first reading, then compare with a reference thermometer/hygrometer; note any constant offset.
- **Vibration pot:** Adjust so idle is solid LOW and gentle taps produce clean, countable pulses.
- **Record:** All offsets, scale factors and pot settings in code comments and the report.

## 15. Test Plan

| ID | Test Case | Expected Result |
|---|---|---|
| T1 | Power-on, sensors idle | NORMAL, valid readings, no alarm |
| T2 | Vibration HIGH > 2 s or > 3 pulses in 2 s | WARNING within confirmation window + 1 s; alarm on |
| T3 | Sustained vibration intensities | CRITICAL; continuous/fast alarm |
| T4 | Temperature > 45 C for 3 s (safe heat) | WARNING then CRITICAL per thresholds |
| T5 | Current > 1.5 A then > 2.5 A | WARNING / CRITICAL as configured |
| T6 | Acknowledge while WARNING | Buzzer off; visual WARNING retained; SILENCED=1; LED still WARNING pattern |
| T7 | While SILENCED in WARNING, escalate to CRITICAL | SILENCED cleared; buzzer re-enabled (new ack required); LED CRITICAL pattern |
| T8 | Fault removed; values below exit thresholds for 5 s | Returns to NORMAL; SILENCED cleared |
| T9 | DHT22 read interval under normal conditions | >= 2 s (impl. >= 2.5 s); no NaN |
| T9b | DHT22 disconnected / data line open | Invalid/NaN detected (FR10); error on OLED & dashboard; other sensors continue |
| T10 | Apply 5.0 V at divider input (bench supply) | ADC pin voltage ~3.0 V and < 3.1 V |
| T11 | 3-4 laptops on SoftAP + dashboard | All update; latency < 1.5 s; if unstable keep only Laptop 2 |


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| ID | Test Case | Expected Result |
|---|---|---|
| T12 | 45 min continuous run + artificial hang (debug flag) | No unrecoverable hang; watchdog (FR9) recovers within a few seconds |
| T13 | Any other sensor disconnect | Error flag on OLED/dashboard; remaining sensors continue (FR10) |
| T14 | LED/buzzer/SILENCED sequence | Ack in WARNING: buzzer off, LED still WARNING; escalate to CRITICAL: buzzer re-enables, LED CRITICAL |
| T15 | ACS712 zero calibration (no load) | After applying zero_offset, reported current is approximately 0 A |

## 16. Project Timeline & Milestones

| Phase | Focus | Activities |
|---|---|---|
| Week 1-2 | Hardware | BOM, 3.3 V power for digital sensors, ACS712 at 5 V + divider, individual sensor bring-up |
| Week 3-4 | Firmware core | Scheduler (>=2.5 s DHT), filters, debounce, FSM + SILENCED, OLED, watchdog FR9, invalid detection FR10 |
| Week 5 | Web + integration | SoftAP WPA2, SSE dashboard, Serial/LittleFS logging, multi-client test |
| Week 6 | Test & demo | Full test plan (incl. T14/T15), calibration, multi-laptop dry-runs, failure-recovery practice, docs |


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## 17. Multi-Laptop Demonstration + Failure Recovery

### Pre-Demo Checklist

- Laptop 1: USB to ESP32, Serial Monitor + Plotter live.
- Laptop 2 (primary dashboard) + Laptop 3/4: connected to SoftAP, dashboard open and updating.
- Printed card: SSID, WPA2 password, IP. OLED also shows password at boot.
- Fault tools ready (tap, safe heat, small motor/fan for current).
- Power bank charged; acknowledge button debounced and tested.
- Watchdog and non-blocking behaviour verified earlier.

### Live Script (8-12 min)

1. **Intro (~1 min):** Problem + sensors.
2. **Normal operation (~2 min):** OLED + Serial + dashboards.
3. **Fault injection (~4-5 min):** Vibration, temperature, current; show WARNING/CRITICAL; demonstrate acknowledge then escalation (SILENCED cleared, buzzer returns).
4. **Technical points (~2 min):** 3.3 V safety, ACS712 at 5 V, divider, ADC formula, safe GPIOs, DHT timing, FSM + SILENCED rule, watchdog.
5. **Close + Q&A (~1-2 min):** Cost, extensions, invite faculty to interact.

### Roles

**A:** Serial/monitoring (Laptop 1). **B:** Fault injection. **C:** Explanation. **D (optional):** Docs / extra dashboard.

### Demo Failure Recovery Checklist

- **Wi-Fi / SoftAP unstable:** Disconnect Laptop 3/4; keep only Laptop 2 as primary dashboard.
- **Complete Wi-Fi failure:** Fall back immediately to OLED + Serial Plotter; core logic remains demonstrable.
- **ESP32 hang:** Watchdog (FR9) should recover within a few seconds. If not, power-cycle and resume.
- **Sensor NaN / disconnect:** Point out the FR10 error flag; continue with remaining sensors.
- **Acknowledge not responding:** Check GPIO 27 wiring and debounce. Any Serial override is debug-only and must not be relied on in the graded demo.
- **Power brown-out:** Switch to a known-good power bank and restart.

## 18. Success Criteria

1. All sensors produce live, sensible readings; DHT22 interval >= 2 s (impl. >= 2.5 s) with no NaN under normal conditions.
2. State transition occurs within the configured confirmation window + 1 s.
3. Acknowledge silences buzzer within 0.5 s while visual state is retained (SILENCED); LED pattern remains driven by state.
4. Escalation WARNING -> CRITICAL while SILENCED clears SILENCED and re-enables the buzzer.
5. Recovery to NORMAL occurs after the exit hysteresis time (~5 s); SILENCED clears.
6. 3-4 laptops can display the dashboard; if unstable, at least Laptop 2 remains solid with latency < 1.5 s.
7. 45 min continuous run without unrecoverable hang (watchdog FR9 present and tested).
8. With 5.0 V applied at the divider input, ADC pin measures ~3.0 V and stays < 3.1 V.
9. Invalid/disconnected sensor is detected and indicated (FR10).


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10. After zero calibration, no-load current reads approximately 0 A (T15).
11. Team can explain 3.3 V logic safety, ACS712 at 5 V + divider, ADC formula, safe GPIOs, DHT timing, non-blocking design, and the SILENCED escalation rule.
12. Documented extra cost is within target (hard limit Rs. 2000).

## 19. Deliverables

- Working prototype with correct 3.3 V digital-sensor power, ACS712 at 5 V, divider and pull-ups.
- Source code (modular, non-blocking, watchdog FR9, invalid detection FR10, commented) + dashboard files.
- This PRD v6.0 Final Freeze.
- Complete block diagram and FSM flowchart (see Appendix; final drawings in report).
- Exact BOM with resistor values, calibration record, completed test-plan results (incl. T14/T15).
- Short report/slides on industrial relevance and microprocessor concepts demonstrated.
- Demo script + pre-demo checklist + failure-recovery checklist.

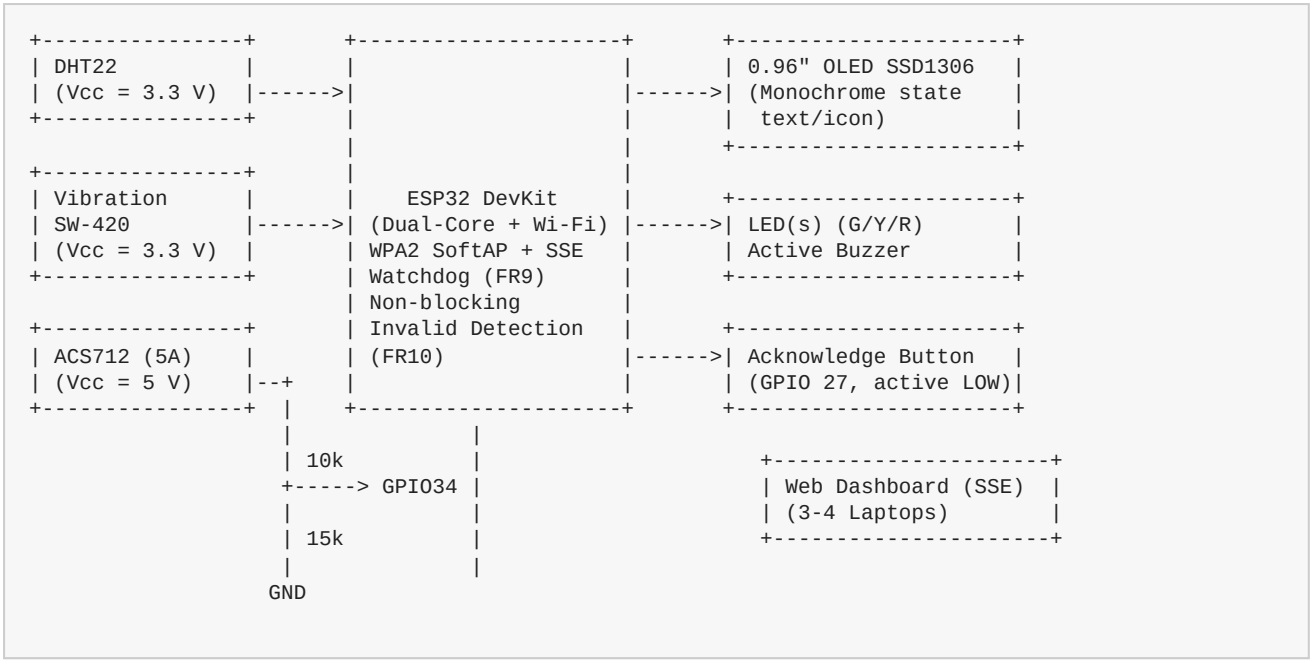
## 20. Future Extensions

- MQTT to free broker + Node-RED.
- Adaptive thresholds or simple on-device anomaly scoring.
- Multi-node gateway.
- Battery + proper power management.
- Industrial enclosure and CT for higher-power motors.

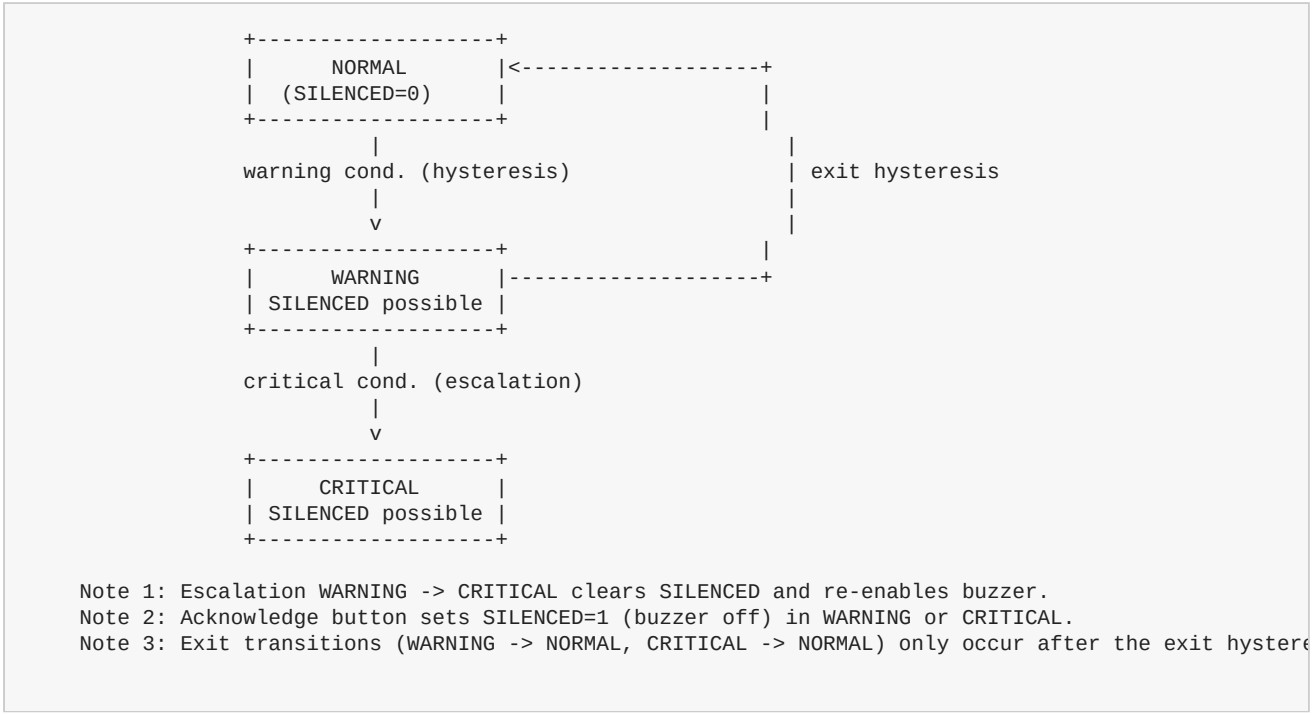

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## Appendix: Complete Block Diagram & FSM Diagram

### A. Complete System Block Diagram




### B. FSM State Diagram





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